

## Freeze ablation – results

Three runs sharing the same setup (HCP-only, T=1200, temporal\_kernel=20, base\_lr=3e-4, effective batch 16, 20 epochs, DINOv2 ViT-S/14 reg4 ImageNet init). Only the parameter-freezing policy differs.

### Strategies

| Strategy                       | Trained components                            | Trainable params |
|--------------------------------|-----------------------------------------------|------------------|
| <b>A – Full fine-tune</b>      | full backbone + heads                         | ~22 M            |
| <b>B – Freeze last3</b>        | patch_embed.* +<br>blocks.9–11 + norm + heads | ~3.6 M           |
| <b>C – Freeze except input</b> | patch_embed.* + heads                         | ~300 k           |

Best checkpoint per strategy (selected among available iters: 5999/8999 for A, 11999/17999 for B and C).

### HCP probes

| Label         | Metric | A (full FT) | B (freeze last3) | C (freeze fmri) |
|---------------|--------|-------------|------------------|-----------------|
| Sex           | AUC    | 0.751       | 0.696            | <b>0.844</b>    |
| BrainVol      | MAE    | 116,179     | 117,744          | <b>99,455</b>   |
| GrayMatterVol | MAE    | 34,762      | 34,815           | <b>29,274</b>   |
| Age           | MAE    | 3.044       | 3.056            | <b>2.931</b>    |
| FluidIntel    | MAE    | 4.047       | 4.035            | <b>4.003</b>    |
| ProcSpeed     | MAE    | 16.343      | 16.401           | <b>16.207</b>   |
| WorkingMem    | MAE    | 10.442      | <b>10.410</b>    | 10.411          |

### ADNI probes

| Label         | Metric | A (full FT)  | B (freeze last3) | C (freeze fmri) |
|---------------|--------|--------------|------------------|-----------------|
| Sex           | AUC    | 0.653        | 0.696            | <b>0.813</b>    |
| CDR           | AUC    | <b>0.542</b> | 0.522            | 0.532           |
| Degradation1Y | AUC    | <b>0.581</b> | 0.543            | 0.559           |
| Degradation2Y | AUC    | <b>0.533</b> | 0.456            | <b>0.533</b>    |
| Degradation3Y | AUC    | 0.528        | 0.468            | <b>0.544</b>    |
| MMSE          | MAE    | 2.478        | <b>2.426</b>     | 2.750           |
| Age           | MAE    | 5.113        | <b>4.993</b>     | 5.703           |

**Bold** = best value in row. AUC: higher is better. MAE: lower is better.

\*Source data: outputs/probes/probe\_{adni,hcp}<run\_name>\_iter<iter>.json – generated by slurm\_jobs/probe\_{adni,hcp}<run\_name>\_iter<iter>.py  
aggregated by python3 scripts/summarize\_probes.py.\*